



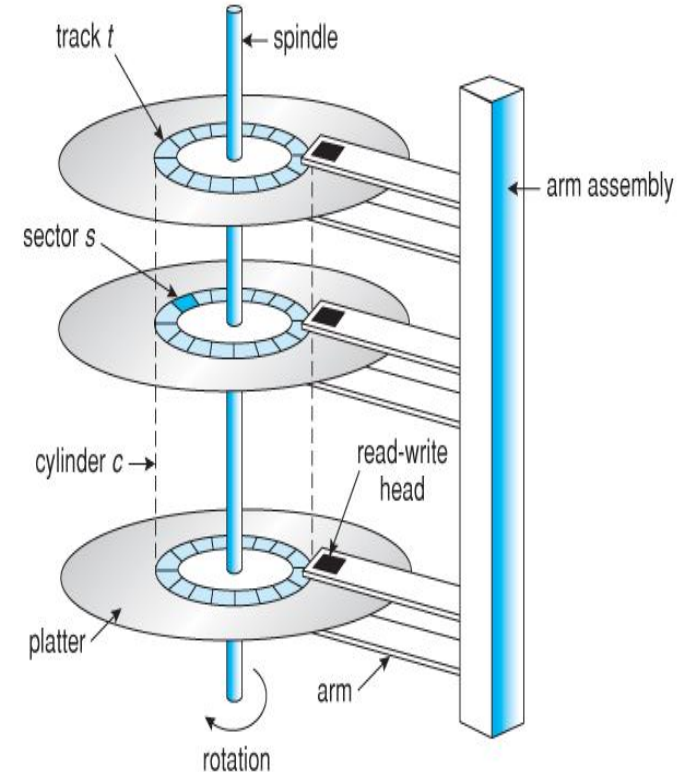
Mass-Storage Structure-I

Dr. Jit Mukherjee



Magnetic Disks

- One or more *platters* (disks) covered with magnetic media.
 - a. *Hard disk* platters -> rigid metal, "floppy" disks -> flexible plastic.
- Each platter has two working *surfaces*.
 - a. Each working surface -> concentric rings called *tracks*.
 - b. The collection of all tracks that are the same distance from the edge of the platter -> *cylinder*.
- Each track -> *sectors*, traditionally containing 512 bytes of data.
 - a. Modern disks occasionally use larger sector sizes.
 - b. Sectors include a header and a trailer, including checksum information among other things.
 - c. Larger sector sizes reduce the fraction of the disk consumed by headers and trailers, but increase internal fragmentation.
- Data on a hard drive is read by *read-write heads*.
 - a. one head per surface, each on a separate *arm*, and controlled by a common *arm assembly* which moves all heads simultaneously from one cylinder to another.
- The storage capacity -> number of heads (number of working surfaces) * number of tracks per surface * number of sectors per track * number of bytes per sector.
- *Positioning/ seek/ random access time* -> to move the heads from one cylinder to another, and for the heads to settle down. Slowest
- *Rotational latency* -> time required for find desired sector by rotating.
- *transfer rate* -> move the data electronically from the disk to the computer
- Disk heads "fly" over surface on a very thin cushion of air -> *Head Crush*



Solid State Disks and Magnetic Tapes

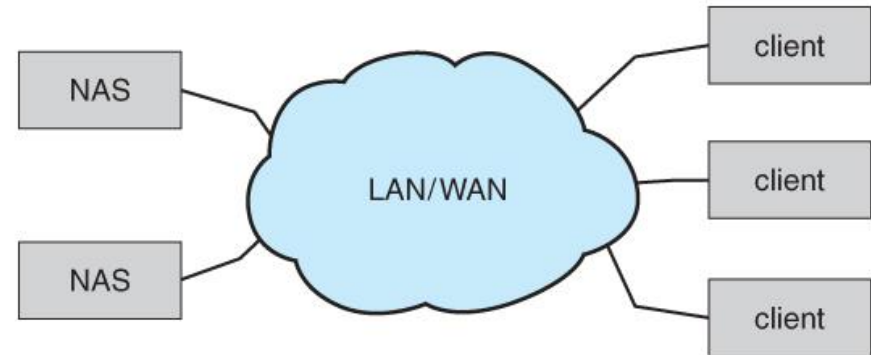
- Floppy -> *removable*. Hard drives -> removable, even *hot-swappable*, (can be removed while computer is running).
- Disk drives are connected to the computer via a cable known as the *I/O Bus*
 - *Enhanced Integrated Drive Electronics (EIDE); Advanced Technology Attachment (ATA); Serial ATA (SATA), Universal Serial Bus (USB); Fiber Channel(FC), and Small Computer Systems Interface (SCSI).*
- The *host controller* -> computer end of the I/O bus, and the *disk controller* is built into the disk itself.
- Magnetic Tapes
 - Once commonly used secondary storage before hard disk drives, today primarily for backups.
 - Accessing a particular spot on a magnetic tape can be slow, but once reading or writing commences, access speeds are comparable to disk drives
 - Capacities of tape drives -> 20 to 200 GB,
- Solid-State Disks - SSDs use memory technology as a small fast hard disk.
 - Flash memory or DRAM chips protected by a battery to sustain the information through power cycles.
 - No moving parts -> much faster. expensive.
 - High-speed cache of hard-disk information that must be accessed quickly.

Disk Structure

- Addressed as large one-dimensional arrays of logical blocks -> 512 bytes
- This logical blocks are mapped onto the sector sequentially
 - Sector 0 is the first sector of the first track on the outermost cylinder
 - Mapping ordered through track -> rest of the tracks of the cylinder -> rest of the cylinder from outermost to innermost
- Logical block -> cylinder number, track number, sector number
 - Issues with defective sectors
- With *Constant Linear Velocity, CLV*, the density of bits is uniform from cylinder to cylinder. Because there are more sectors in outer cylinders, the disk spins slower when reading those cylinders, causing the rate of bits passing under the read-write head to remain constant.
 - This is the approach used by modern CDs and DVDs.
- With *Constant Angular Velocity, CAV*, the disk rotates at a constant angular speed, with the bit density decreasing on outer cylinders.
 - These disks would have a constant number of sectors per track on all cylinders.

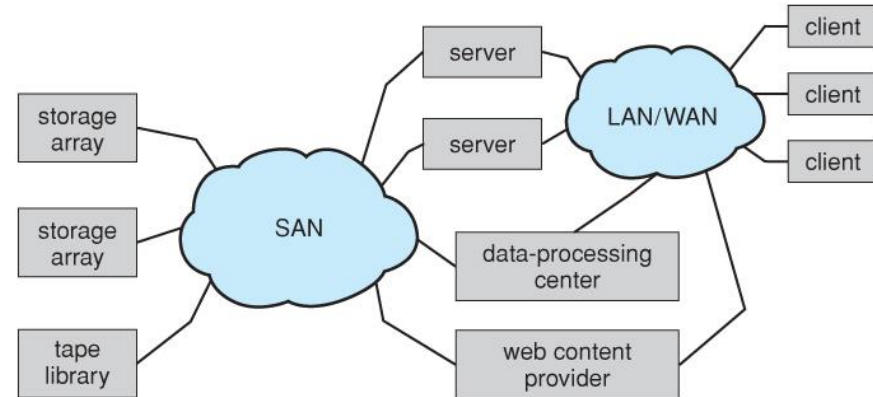
Disk Attachment: Host Attached Stores

- Computer access disk storage -> Via I/O port; host attached storage or Via a remote host; network attached storage
- Host Attached Storage - Access through local I/O port -> typically IDE / ATA
 - Maximum two drives per I/O
 - High end architecture -> SCSI or Fiber Channel (FC)
- SCSI -> supports 16 devices on the bus -> SCSI Initiator and SCSI Targets
- FC -> High speed serial architecture operate over optical fibre
- Network attached storage connects storage devices to computers using a remote procedure call, RPC, interface, typically with something like NFS filesystem mounts.



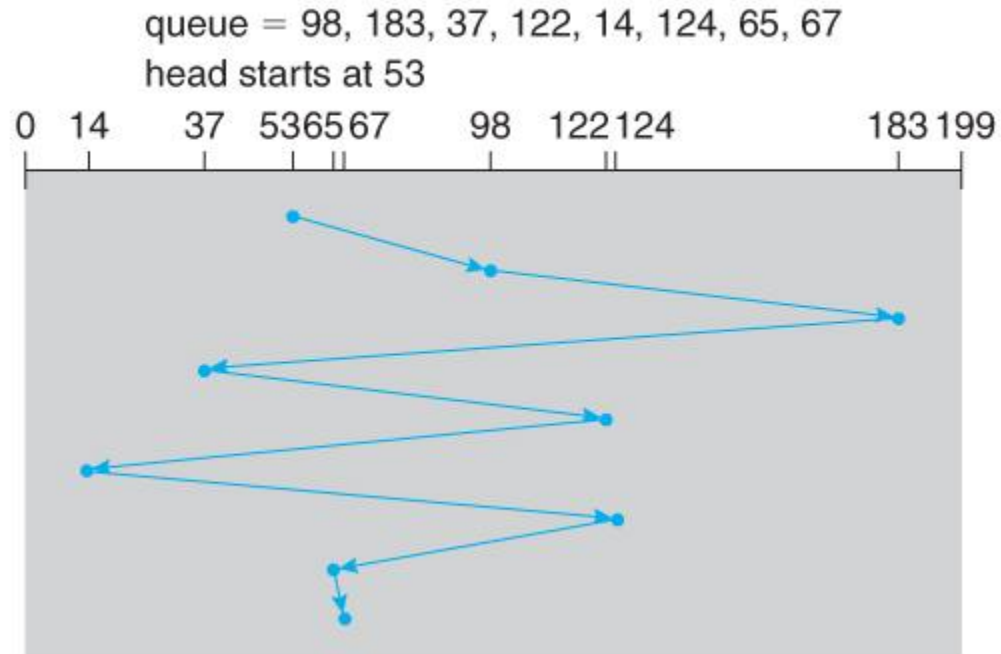
Network Attached Stores & Storage Area Network

- This is convenient for allowing several computers in a group common access and naming conventions for shared storage.
 - NAS can be implemented using SCSI cabling, or *iSCSI* uses Internet protocols and standard network connections, allowing long-distance remote access to shared files.
 - NAS allows computers to easily share data storage, but tends to be less efficient than standard host-attached storage.
-
- *Storage-Area Network, SAN*, connects computers and storage devices in a network, using storage protocols instead of network protocols.
 - SAN is very flexible and dynamic, allowing hosts and devices to attach and detach on the fly.
 - SAN is also controllable, allowing restricted access to certain hosts and devices



Disk Scheduling & FCFS

- Disk transfer speeds are limited primarily by *seek times* and *rotational latency*.
 - When multiple requests are to be processed there is also some inherent delay in waiting for other requests to be processed.
- *Bandwidth* is measured by the amount of data transferred divided by the total amount of time from the first request being made to the last transfer being completed
- Both bandwidth and access time can be improved by processing requests in a good order.
- Disk requests include the disk address, memory address, number of sectors to transfer, and whether the request is for reading or writing.
- **First-Come First-Serve** is simple and intrinsically fair, but not very efficient. Consider in the following sequence the wild swing from cylinder 122 to 14 and then back to 124:
- Total Head Movement 640 cylinder

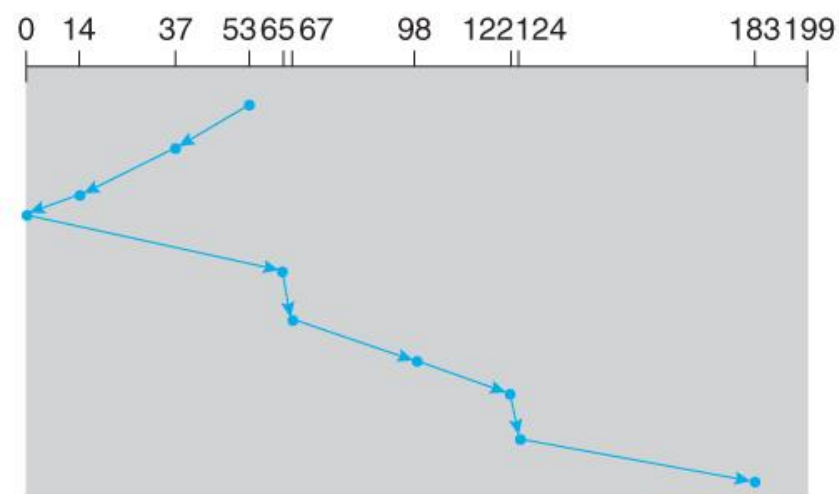


SSTF and SCAN Scheduling

- **Shortest Seek Time First scheduling** is more efficient, but may lead to starvation if a constant stream of requests arrives for the same general area of the disk -> Problem similar with Shortest Job First
- SSTF reduces the total head movement to 236 cylinders, down from 640 required for the same set of requests under FCFS. Note, however that the distance could be reduced still further to 208 by starting with 37 and then 14 first
- The **SCAN algorithm (elevator algorithm)** moves back and forth from one end of the disk to the other, similarly to an elevator processing requests in a tall building.
- Under the SCAN algorithm, If a request arrives just ahead of the moving head then it will be processed right away, but if it arrives just after the head has passed, then it will have to wait for the head to pass going the other way on the return trip. This leads to a fairly wide variation in access times which can be improved upon.

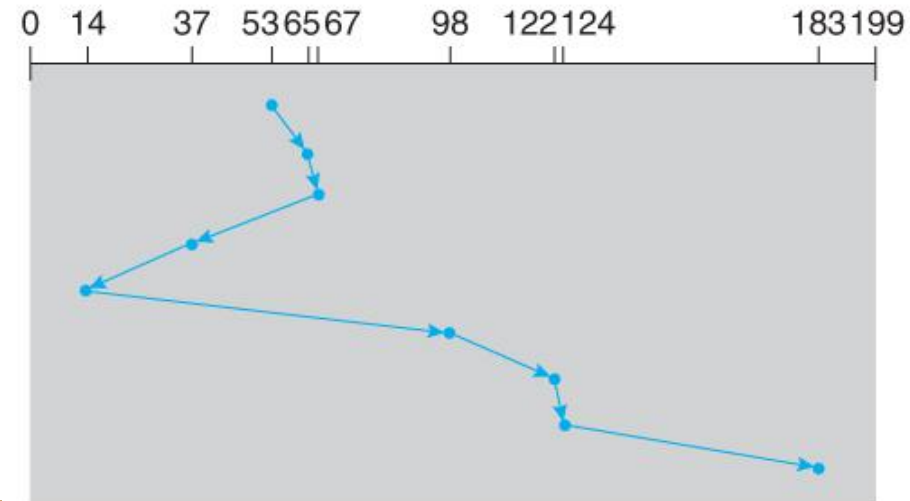
queue = 98, 183, 37, 122, 14, 124, 65, 67

head starts at 53



queue = 98, 183, 37, 122, 14, 124, 65, 67

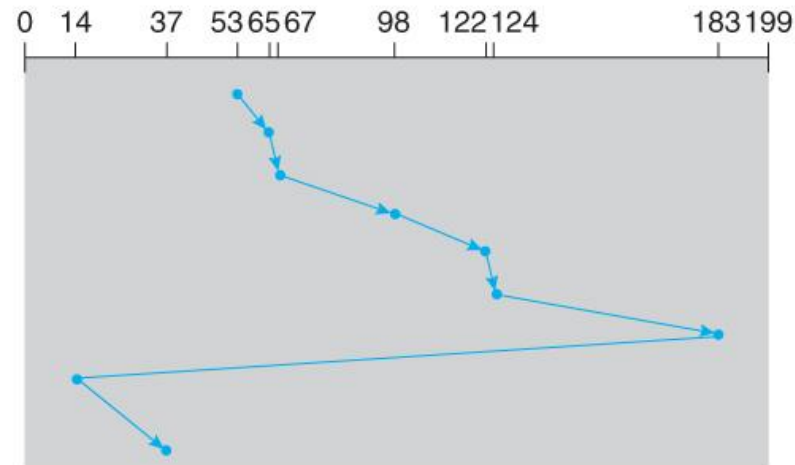
head starts at 53



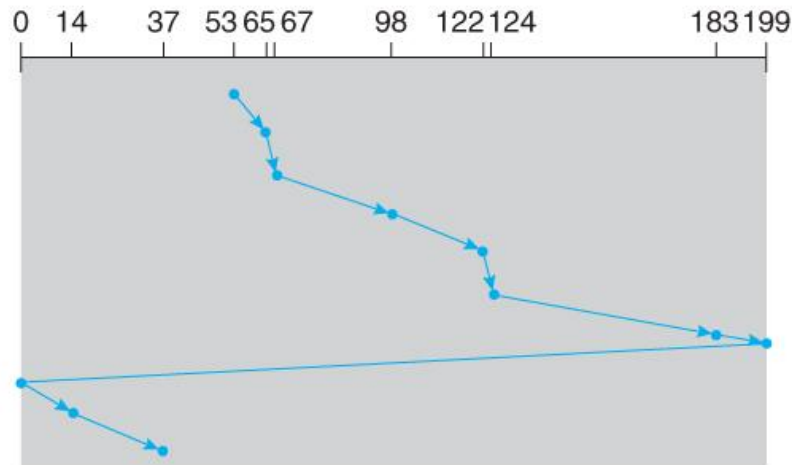
C-SCAN and Look Scheduling

- The *Circular-SCAN* algorithm improves SCAN by treating all requests in a circular queue fashion - Once the head reaches the end of the disk, it returns to the other end without processing any requests, and then starts again from the beginning of the disk:
- *LOOK* scheduling improves upon SCAN by looking ahead at the queue of pending requests, and not moving the heads any farther towards the end of the disk than is necessary.
- Look and C-Look scheduling -> SCAN and C-SCAN

queue = 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53



queue = 98, 183, 37, 122, 14, 124, 65, 67
head starts at 53



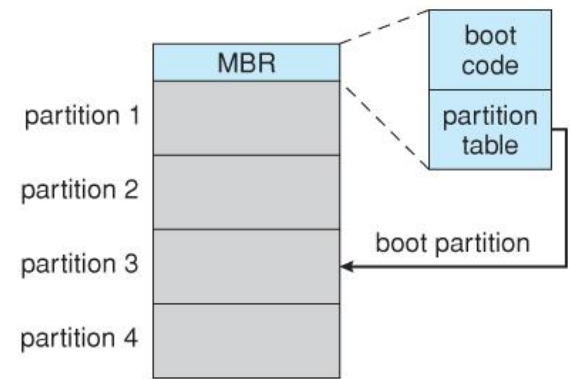
Selection of Disk Scheduling Algorithm

- With very low loads all algorithms are equal, since there will normally only be FCFS.
- For slightly larger loads, SSTF offers better performance than FCFS, but may lead to starvation.
- For busier systems, SCAN and LOOK algorithms eliminate starvation problems.
- The actual optimal algorithm may be something even more complex.
- Some improvement to overall filesystem access times can be made by intelligent placement of directory. If those structures are placed in the middle of the disk instead of at the beginning of the disk, then the maximum distance from those structures to data blocks is reduced to one-half of the disk size.
 - If structures have their data blocks stored as close as possible to the corresponding directory structures, then that reduces still further the overall time to find the disk block numbers and then access the corresponding data blocks.
- On modern disks the rotational latency can be as significant as the seek time

Disk Management: Disk Formatting

- Before a disk can be used, it has to be **low-level formatted** -> mark all of the headers and trailers as beginning and ends of each sector. Included in the header and trailer -> linear sector numbers *error-correcting codes, ECC*,
 - To allow damaged sectors to be detected, and recovered (depending on the extent of the damage.) Sector sizes -> 512 bytes, but may be larger, particularly in larger drives.
- ECC calculation is performed with every disk read or write
 - if damage is detected but the data is recoverable, then a **soft error** has occurred.
 - Soft errors are generally handled by the on-board disk controller, and never seen by the OS.
- Once the disk is low-level formatted -> partition the drive into one or more separate partitions.
 - This step must be completed even if the disk is to be used as a single large partition so that the partition table can be written to the beginning of the disk.
- After partitioning filesystems -> **logically formatted** -> Stores initial file-system data structure into disk -> Include maps of free and allocated spaces
 - Clusters - Groups blocks together into larger chunks

Disk Management: Boot Blocks



- ROM contains a **bootstrap** program (OS independent) -> enough code to find the first sector on the first hard drive on the first controller, load that sector into memory, and transfer control over to it.
- The first sector on the hard drive -> **Master Boot Record, MBR** -> contains a very small amount of code in addition to the *partition table* -> how the disk is partitioned into logical disks, and indicates which partition is the *active* or *boot* partition.
- Boot program looks -> active partition to find an operating system -> loads more advanced boot program.
- In a *dual-boot* (or larger multi-boot) system, the user may be given a choice of which operating system to boot, with a default action to be taken in the event of no response within some time frame.
- Once the kernel is found by the boot program, it is loaded into memory and then control is transferred over to the OS.
 - a. The kernel will normally continue the boot process by initializing all important kernel data structures, launching important system services (e.g. network daemons, sched, init, etc.), and finally providing one or more login prompts.